

HARIOM KUMAR

AI Engineer | Generative AI & LLM Applications | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI Engineer at Adhyay AI building LLM-powered EdTech features: AI-generated interactive simulations, bilingual app support, and AI video generation. IIT Madras BS Data Science student with diplomas in Data Science and Programming. Skilled in **Python**, **FastAPI**, **Flask**, **React**, **Next.js**, **RAG**, **LangChain**, **FAISS**, **prompt engineering**, **LLM evaluation**, **PostgreSQL**, and **Docker**.

EXPERIENCE

AI Engineer | Adhyay AI Pvt. Ltd.

Sep 2026 – Present

Personalized learning platform for schools (EdTech)

Remote

- Building **Prayog**, an **LLM-powered simulation generator** that turns a class topic into a guided science lesson with an interactive **HTML/JavaScript canvas simulation**, practice questions, and free-explore mode.
- Prototyped a **teacher-student LLM pipeline** (frontier model plans, smaller low-cost model generates) grounded in **NCERT** data from **PostgreSQL**, with an **A/B testing harness** comparing routes on quality and **token cost**.
- Measure every **prompt engineering** and **output validator** change on a fixed **evaluation set** (6 topics × 3 runs).
- Making the existing Adhyay app **bilingual** via **internationalization (i18n)** of its **Next.js + Tailwind CSS** frontend; developing **AI video generation** for lessons.

PROJECTS

Doc Mentor: AI-Powered RAG Chatbot | *Python, FastAPI, LangChain, Gemini, FAISS, Docker, Streamlit*

- Built a **Retrieval-Augmented Generation (RAG)** chatbot for Q&A over uploaded PDF/text documents using **FastAPI**, **LangChain**, **Gemini 3.5 Flash Lite**, Sentence Transformers embeddings, and a **FAISS** vector store.
- Tuned chunk size, **top-k retrieval**, and prompts to **reduce hallucinations**; containerized with **Docker**, API-tested in **Postman**.

Gyantra: AI-Assisted Education Platform | *FastAPI, React, Vite, LLM Orchestration, SSE*

[GitHub](#)

- Built a **10-stage LLM pipeline** converting educational documents into structured Teacher Knowledge Packages, with a **React/Vite** review UI and live progress via **Server-Sent Events** from **FastAPI**.

Kartavya: Civic Complaint Management Platform | *Backend Developer, Team of 5, Scrum*

- Designed a **normalized database schema** (users, zones, complaints, evidence, points, badges, history) and implemented **geo-proximity complaint grouping** with the **Haversine formula** and zone-based routing.
- Built complaint lifecycle workflows, audit trails, and **role-based access control (RBAC)**; used GitHub PR reviews.

Navigating Algorithmic Anxiety: GenAI & Workplace Stress | *Qualitative Research, Team of 5, 2026*

[GitHub](#)

- Co-authored a study on how a **growth mindset** helps professionals handle stress and **skill obsolescence** from **Generative AI**, using **semi-structured interviews**, **thematic analysis**, and a project plan with **WBS** and risk management.

Placement Portal | *Flask, Vue.js, SQLite, REST APIs*

[GitHub](#)

- Built a **full-stack placement platform (Flask REST APIs, Vue.js)** with **RBAC** and company/job approval workflows.

Cinema Audience Forecasting | *Python, Machine Learning, NumPy, Matplotlib*

[GitHub](#)

- Built **machine learning** models to forecast cinema audience demand, using **feature engineering**.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, HTML, CSS

AI / ML: LLMs, Generative AI, RAG, LangChain, Prompt Engineering, LLM Evaluation, Hugging Face, Scikit-Learn

Backend: FastAPI, Flask, REST APIs, SQLAlchemy, Server-Sent Events (SSE), RBAC

Frontend: React, Next.js, Vue.js, Vite, Tailwind CSS, shadcn/ui, Streamlit, Internationalization (i18n)

Databases: PostgreSQL, SQLite, FAISS (Vector Database)

Tools: Git, GitHub, Docker, Postman, OpenRouter, NumPy, Matplotlib, Power BI, Excel

EDUCATION

Indian Institute of Technology Madras | BS in Data Science and Applications

2023 – Present

- Diploma in Programming and Application Development (May 2026); Diploma in Data Science (Dec 2025)
- Coursework:** Machine Learning, Deep Learning, Software Engineering, DBMS, Data Structures & Algorithms

Vikas Vidyalaya, Begusarai | Class XII (CBSE)

2023

CERTIFICATIONS

- Advanced Certificate in Machine Learning & Data Science (IIT Madras); Power BI Dashboards (IIT Madras) ([View](#))
- Advanced Certificate in Programming & Application Development (IIT Madras); SQL Advanced (HackerRank) ([View](#))

ACHIEVEMENTS

- Winner**, IPL Auction Showdown 4.0, IIT Madras Paradox 2026; **Runner-up**, IPL Arena, IIT Madras Paradox 2024
- Participant, DataVista Analytics Competition (Paradox 2025) and TradeQuest 2024